

Tristan Luca Saidi

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EDUCATION

Carnegie Mellon University

Ph.D. Statistics and Data Science, expected June 2029

GPA: 4.00

Pittsburgh, Pennsylvania

Sept 2025–Present

Columbia University

B.S./M.S. in Computer Engineering and Computer Science, *Summa Cum Laude*

GPA: 4.10

New York, New York

Jun 2024

PUBLICATIONS AND PRE-PRINTS

1. T. L. Saidi, G. Mena, F. Gunsilius, L. Wasserman “Second-order Sinkhorn Geometry and $\sqrt{n}$ -Inference”, *in progress*, 2026.
2. T. L. Saidi, A. Kuchibhotla “Locally Linear Generalized Single Index Regression”, *in progress*, 2026.
3. T. L. Saidi, G. Mena, F. Gunsilius, “On the Differential-Geometric Equivalence of Hellinger-Kantorovich and Cone-Wasserstein Spaces”, *preprint*, 2026.
4. T. L. Saidi, G. Mena, L. Wasserman, F. Gunsilius, “Parallel Transport on the Space of Probability Measures for Counterfactual Dynamics Prediction”, *preprint*, 2026.
5. T. L. Saidi, A. Hickok, B. Rieck, A. J. Blumberg, “EmbedOR: Provable Cluster-Preserving Visualizations with Curvature-Based Stochastic Neighbor Embeddings”, *The Proceedings of the National Academy of Sciences*, 2026.
6. T. L. Saidi, A. Hickok, A. J. Blumberg, “Recovering Manifold Structure Using Ollivier-Ricci Curvature”, *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. (Spotlight work, top 5%)
7. G. Guo, T. L. Saidi, M. Terban, M. Valsecchi, S.L. Billinge, H. Lipson, “Ab Initio Structure Solutions from Nanocrystalline Powder Diffraction Data”, *Nature Materials*, 2025.
8. G. Khandate*, T. L. Saidi*, S. Shang*, E. Chang, Y. Liu, J. Adams, M. Ciocarlie, “R×R: Rapid eXploration for Reinforcement Learning via Sampling-based Reset Distributions and Imitation Pre-training”, *RSS Special Issue: Autonomous Robots*, 2024.
9. G. Khandate, S. Shang, E. Chang, T. L. Saidi, Y. Liu, S. Dennis, J. Adams, M. Ciocarlie, “Sampling-based Exploration for Reinforcement Learning of Dexterous Manipulation”, *19th Robotics: Science and Systems (RSS)*, Daegu, Republic of Korea, 2023.

* denotes equal contribution.

SCHOLAR METRICS

Google Scholar Citations: 136

h-index: 4

i10-index: 2

WORK EXPERIENCE

Citadel Securities

Quantitative Researcher Intern

Miami, Florida

Jun 2026–Aug 2026

- Interned at Citadel Securities as a Quantitative Researcher on the options team.

AffectusAI

Machine Learning Engineer

New York, New York

July 2024–Nov 2024

- Designed scalable manifold learning pipelines in Python, and led the development of a data visualization pipeline with Pinecone Vector DB and the OpenAI API.

RESEARCH EXPERIENCE

Department of Statistics and Data Science

Research Assistant, Department of Statistics and Data Science

Pittsburgh, Pennsylvania

Sept 2025–Present

- Developing the theory of parallel transport on the Wasserstein manifold for applications in Causal Inference with Professor Larry Wasserman, Gonzalo Mena and Florian Gunsilius.
- Conducting research at the intersection of semi-parametric statistics, geometry and machine learning with Professor Arun Kuchibhotla.

Andrew Blumberg Group

Research Assistant, Departments of Computer Science and Mathematics

New York, New York

Jan 2024–Aug 2025

- Led the development of a scalable stochastic neighbor embedding algorithm in Python using CVXPY and Cython, the paper for which is published at the Proceedings of the National Academy of Sciences.
- Implemented an efficient Python graph pruning algorithm with an accessible API for data scientists using multiprocessing and Cython. This work was published at as a spotlight work at ICLR 2025.

Robotic Manipulation and Mobility (ROAM) Lab

Research Assistant, Department of Computer Science

New York, New York

Jan 2022–May 2024

- Investigated sample-efficient reinforcement learning techniques for dexterous robotic control using both real-world and simulated environments using Python, Pytorch, and C++.

Bioelectronic Systems Laboratory

Research Assistant, Department of Electrical Engineering

New York, New York

Jan 2021–Jan 2022

- Worked with optimization concepts to derive and help write the proof of convergence for the least squares cost function for a novel compressed sensing algorithm. Also implemented MATLAB code to generate figures and statistics for error and accuracy.

TEACHING EXPERIENCE

Carnegie Mellon University

Teaching Assistant, Department of Statistics and Data Science

Pittsburgh, Pennsylvania

August 2025 – Present

- Head Teaching Assistant for *Probability and Mathematical Statistics*, *Statistical Computing*, and *Statistical Machine Learning*.

Columbia University Department of Computer Science

Teaching Assistant, Department of Computer Science

New York, New York

Jun 2022–Sep 2022

- Assisted in teaching a graduate Machine Learning class taught by Professor Nakul Verma. Course covered topics such as SVMs, Kernel methods, KNN, Dimensionality Reduction, Clustering, and more.

HONORS AND AWARDS

Computer Engineering Award of Excellence

Jun 2023

Summa Cum Laude Graduate from the Columbia School of Engineering and Applied Sciences

Jun 2023

Columbia Engineering Alumni Association (CEAA) Scholar

Aug 2022

Columbia University Dean's List: Fall 2020, Spring 2021, Fall 2021, Spring 2022, Fall 2022

RELEVANT COURSES

Carnegie Mellon University: Advanced Statistical Theory, Intermediate Statistics, Regression Analysis, Statistical Optimal Transport, Measure Theory

Columbia University: Geometric Data Analysis, High Dimensional Probability, Machine Learning, Convex Optimization, Foundations of Graphical Models, Deep Learning, Reinforcement Learning, Operating Systems

SKILLS

Proficient: Python, R, PyTorch, PyTorch Geometric, Git, NumPy, Pandas, Scikit-Learn, L^AT_EX, Matplotlib, Linux (Usage and Shell Scripting), IsaacGym, Mujoco, C

Familiar: C++, Java, SciPy, Matlab

PROJECTS

EmbedOR: Cluster-preserving visualizations with curvature-based Stochastic Neighbor Embeddings	Aug 2025
ORC-ManL: Recovering manifold structure using Ollivier-Ricci Curvature	Oct 2024
Scalar Curvature Estimation with Graph Neural Networks	Mar 2024
Exploring Adversarial Perturbations to Diffusion Models	Dec 2023
Examining the geometry of neural mode connecting loss subspaces	Jun 2023
Implementation of a functional scheduler and filesystem for Linux 5.10.182	June 2023
On the Effect of Unsupervised Regularization for Image Classification	Dec 2022
Implementation of a Finite Impulse Response (FIR) filter in Verilog	Dec 2021

RESEARCH STUDENTS SUPERVISED

Geoffrey Wu , Columbia B.S. Computer Science	Jan 2024–Jun 2024
Jamie Palmer , Columbia M.S. Mechanical Engineering	Jan 2024–Jun 2024
Ethan Matzner , Columbia B.S. Mechanical Engineering	Sep 2023–Jun 2024

SELECTED PRESS

Time Magazine Best Inventions of 2023	Time Magazine
“Robot Hand Manipulates Complex Objects by Touch Alone”	IEEE Spectrum
“This robot hand can manipulate objects without seeing them”	The Robot Report
“Highly dexterous robot hand can operate in the dark—just like us”	TechXplore
“This agile robotic hand can handle objects just by touch”	Popular Science

SERVICE

Co-organizer of Carnegie Mellon University’s <i>Optimal Transport reading group</i>	Sep 2026
Reviewer for 2025 <i>International Conference on Learning Representations (ICLR)</i>	Oct 2024
Volunteer for <i>Robotics: Science and Systems (RSS)</i> Conference	Jun 2022